

Generated MCQs

Question 1 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

How many edges does a hexagonal prism have?

- A. 18
- B. 12
- C. 16
- D. 24

Question 2 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

If the base of a prism is an n -sided polygon, how many edges does the prism have?

- A. It has $3n$ edges.
- B. It has $2n$ edges.
- C. It has $n + 2$ edges.
- D. It has $4n$ edges.

Question 3 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

How many vertices does a pyramid have if the base is an n -sided polygon?

- A. $n + 1$
- B. $2n$
- C. n
- D. $n + 2$

Question 4 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

Identify the prisms that have exactly 7 vertices.

- A. *There are no such prisms.*
- B. *Pentagonal prisms.*
- C. *Triangular prisms.*
- D. *Heptagonal prisms.*

Question 5 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 1)

Identify the true statement about a pentagonal prism.

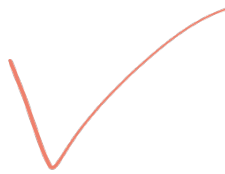
- A. *A pentagonal prism has seven faces.*
- B. *A pentagonal prism has six faces.*
- C. *A pentagonal prism has ten faces.*
- D. *A pentagonal prism has eight faces.*



Question 6 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

How many diagonals does a hexagon have?

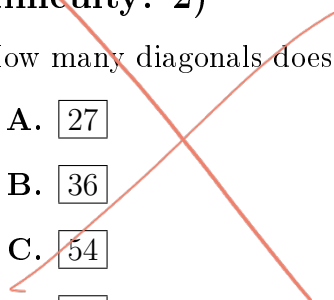
- A. $\frac{6 \times (6-3)}{2} = 9$
- B. $\frac{6 \times (6-2)}{2} = 12$
- C. $6 \times (6-3) = 18$
- D. $6 \times (6-1) = 30$



Question 7 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

How many diagonals does a regular nonagon have?

- A.
- B.
- C.
- D.



Question 8 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

What is the sum of the amplitudes of the exterior angles of a triangle?

- A. 360°
- B. 180°
- C. 270°
- D. 540°



Question 9 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

A regular decagon (10-sided polygon) has how many diagonals?

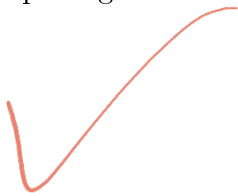
- A. 35 diagonals
- B. 45 diagonals
- C. 70 diagonals
- D. 40 diagonals



Question 10 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

A prism with a pentagonal base has 7 faces and 15 edges. How many vertices does it have?

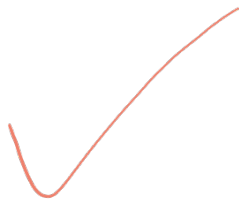
- A. 10
- B. 6
- C. 8
- D. 20



Question 11 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 4)

Which of the following correctly lists the number of faces, vertices, and edges for a heptagonal prism?

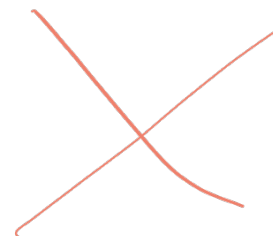
- A. 9, 14, 21
- B. 8, 14, 21
- C. 9, 15, 21
- D. 9, 14, 20



Question 12 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 4)

Which Platonic solid has twice as many edges as vertices?

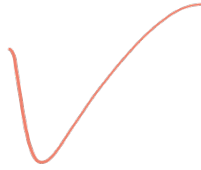
- A. Octahedron
- B. Cube
- C. Dodecahedron
- D. Icosahedron



Question 13 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

A solid has a total of 16 faces and edges. Which of the following is the solid?

- A. *The solid is a pentagonal pyramid.*
- B. *The solid is a hexagonal pyramid.*
- C. *The solid is a pentagonal prism.*
- D. *The solid is a hexagonal prism.*



Question 14 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

What is the number of edges in a hexagonal prism?

- A. 18
- B. 12
- C. 8
- D. 24



Question 15 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

A prism with a heptagonal base has how many edges?

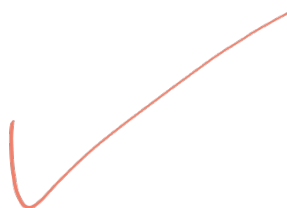
- A.
- B.
- C.
- D.



Question 16 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

The measure of an exterior angle of a regular pentagon is

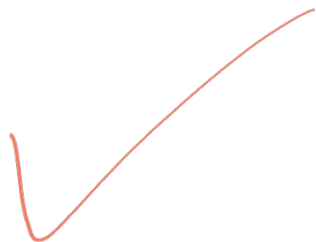
- A. 72°
- B. 108°
- C. 60°
- D. 36°



Question 17 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

The measure of an interior angle of a regular hexagon is

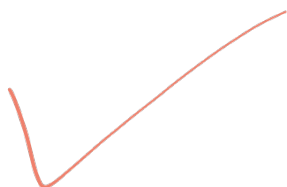
- A. 120°
- B. 150°
- C. 60°
- D. 90°



Question 18 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

A polygon has 54 diagonals. How many sides does it have?

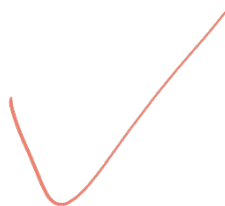
- A. 12
- B. 10
- C. 15
- D. 9



Question 19 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

If a prism has a base that is a heptagon, how many faces does it have?

- A. 9
- B. 8
- C. 14
- D. 10



Question 20 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

Which statement is true about the exterior angles of a triangle?

- A. The sum of the measures of the exterior angles is 360° .
- B. The sum of the measures of the exterior angles is 180° .
- C. *Each exterior angle is equal to the sum of the two adjacent interior angles.*
- D. *The sum of the measures of the exterior angles depends on the type of triangle.*



Question 21 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

Consider a triangle with angles measuring 40° , 60° , and 80° . Choose the true statement.

- A. *It is an acute triangle.*
 - B. *It has a right angle.*
 - C. *It is an obtuse triangle.*
 - D. *It is an equilateral triangle.*
- 

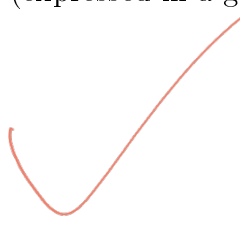
Question 22 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

Choose the true statement about triangle congruence.

- A. *If two triangles have all three sides congruent, then they are congruent.*
 - B. *If two triangles have two angles congruent, then they are congruent.*
 - C. *If two triangles have two sides and a non-included angle congruent, then they are congruent.*
 - D. *If two triangles have one side and one angle congruent, then they are congruent.*
- 

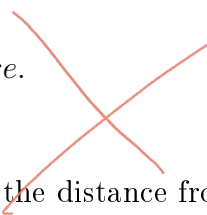
Question 23 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

The total surface area (expressed in a given measure unit) of this cone with radius a and slant height b is:

- A. $\pi a(a + b)$
 - B. πab
 - C. $2\pi a(a + b)$
 - D. $\pi a^2 + \pi b^2$
- 

Question 24 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

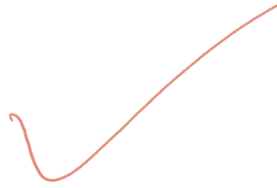
Consider a reflection over line s . Choose the correct statement.

- A. *The original figure and its image are congruent.*
 - B. *The image is similar but not congruent to the original figure.*
 - C. *The line s must be horizontal.*
 - D. *The image is located on the opposite side of s at twice the distance from s .*
- 

Question 25 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

Which geometric transformation is an isometry that maps every line to a parallel line?

- A. *Translation*
- B. *Rotation*
- C. *Reflection*
- D. *Glidereflection*



Question 26 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 4)

Which of the following geometric transformations changes the size of a figure?

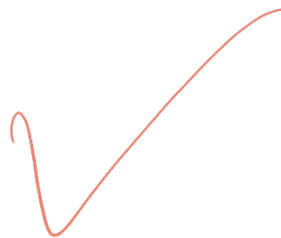
- A. Dilation of centre O and scale factor $k = 3$
- B. Translation associated to vector $\vec{v} = (5, -2)$
- C. Rotation of centre O and amplitude 60°
- D. Reflection associated to line $y = -x$



Question 27 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

In triangle ABC , if the exterior angle at vertex C measures 110° and one of the non-adjacent interior angles measures 45° , what is the measure of the other non-adjacent interior angle?

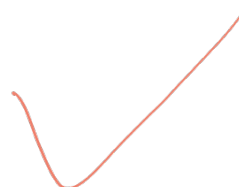
- A. 65°
- B. 45°
- C. 110°
- D. 155°



Question 28 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

Which of the following correctly describes the number of lines of symmetry for a square?

- A. *Asquarehas4linesofsymmetry.*
- B. *Asquarehas2linesofsymmetry.*
- C. *Asquarehas1lineofsymmetry.*
- D. *Asquarehas6linesofsymmetry.*



Question 29 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

A pentagonal prism has how many edges?

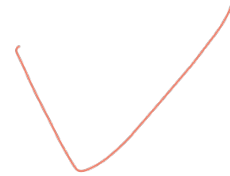
- A. 15
- B. 10
- C. 7
- D. 12



Question 30 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

Which statement correctly describes the rotational symmetries of a regular hexagon?

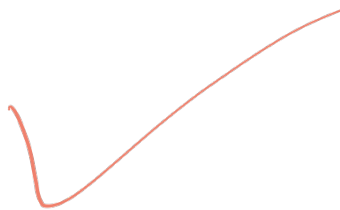
- A. *It has six rotational symmetries, including a 360 rotation.*
- B. *It has three rotational symmetries (60, 120, 180).*
- C. *It has four rotational symmetries (90, 180, 270, 360).*
- D. *It has twelve rotational symmetries (every 30).*



Question 31 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

Which of the following sets of side lengths can form a valid triangle?

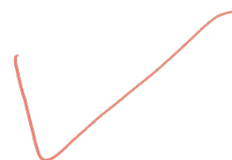
- A. {4, 5, 6}
- B. {3, 4, 8}
- C. {5, 5, 10}
- D. {2, 3, 6}



Question 32 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

Which of the following is a valid criterion for triangle congruence?

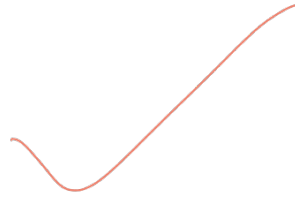
- A. *Two congruent sides and the included angle.*
- B. *Two congruent angles and a non – included side.*
- C. *Three congruent angles.*
- D. *Two congruent sides and a non – included angle.*



Question 33 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

Consider a triangle with sides of length 5 cm, 6 cm, and 7 cm. Which of the following best describes this triangle?

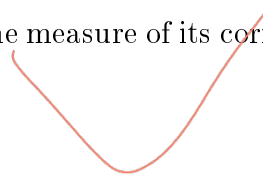
- A. *The triangle is scalene.*
- B. *The triangle is isosceles.*
- C. *The triangle is equilateral.*
- D. *The triangle is right-angled.*



Question 34 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

If an interior angle of a triangle measures 110° , what is the measure of its corresponding exterior angle?

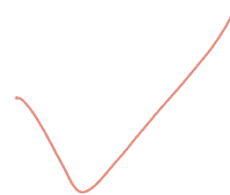
- A. 70°
- B. 110°
- C. 180°
- D. 20°



Question 35 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

In a convex quadrilateral, the diagonals are congruent and bisect each other but are not perpendicular. The quadrilateral is a:

- A. *Rectangle*
- B. *Rhombus*
- C. *Square*
- D. *Parallelogram*



Question 36 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

In a regular hexagon, which statement correctly describes its symmetries?

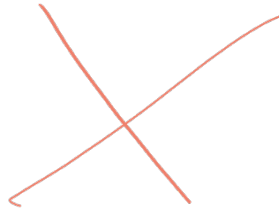
- A. *There are six lines of symmetry and rotational symmetry of order six.*
- B. *There are three lines of symmetry and rotational symmetry of order three.*
- C. *There are six lines of symmetry and rotational symmetry of order three.*
- D. *There are four lines of symmetry and rotational symmetry of order six.*



Question 37 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

A polyhedron has 12 vertices and 30 edges. How many faces does it have?

- A. ☐ 20
- B. ☐ 18
- C. ☐ 22
- D. ☐ 15



Question 38 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

Which of the following is the correct number of diagonals in a regular octagon?

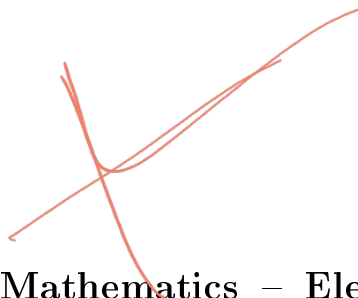
- A. ☐ 20
- B. ☐ 28
- C. ☐ 24
- D. ☐ 40



Question 39 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

How many diagonals does a convex polygon with six sides have?

- A. 9
- B. 15
- C. 12
- D. 18



Question 40 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

What is the formula to calculate the number of diagonals in a polygon with n sides?

- A. $\frac{n(n-3)}{2}$
- B. $\frac{n(n-2)}{2}$
- C. $n(n-3)$
- D. $\frac{n(n+1)}{2}$



Question 41 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

If a prism has a base that is a hexagonal (6-sided) polygon, how many edges does it have?

- A. ☐ 18
- B. ☐ 12
- C. ☐ 8
- D. ☐ 6



Question 42 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

A prism has a base that is a regular hexagon. How many edges does this prism have?

- A. 18
- B. 12
- C. 24
- D. 9



Question 43 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

A hexagonal pyramid has a base with 6 sides. How many faces does it have?

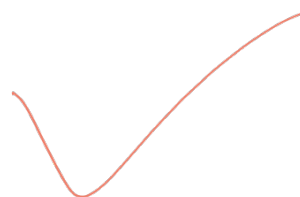
- A. ☐ 7
- B. ☐ 8
- C. ☐ 6
- D. ☐ 12



Question 44 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

In a Cartesian coordinate system, three consecutive vertices A and B of a parallelogram $[ABCD]$ are given, along with the midpoint M of its diagonals. If the coordinates of points A , B , and M are $(0,0)$, $(2,0)$, and $(3,1)$ respectively, what are the coordinates of points C and D ?

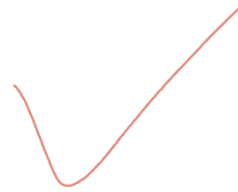
- A. $C(6,2)$ and $D(4,2)$
- B. $C(4,2)$ and $D(6,2)$
- C. $C(6,1)$ and $D(4,1)$
- D. $C(5,2)$ and $D(3,2)$



Question 45 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

If a prism has a pentagonal base, how many edges does it have?

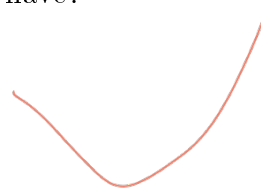
- A. 15
- B. 10
- C. 12
- D. 7



Question 46 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

How many diagonals does a decagon (10-sided polygon) have?

- A. 35
- B. 45
- C. 40
- D. 70



Question 47 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

A convex polygon has 6 sides. How many diagonals does it have?

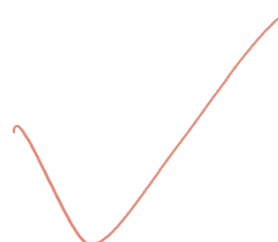
- A. 9
- B. 12
- C. 18
- D. 6



Question 48 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

The sum of the number of edges and vertices of a prism is 30. Choose the right answer.

- A. *The solid is a hexagonal prism.*
- B. *The solid is a pentagonal prism.*
- C. *The solid is a heptagonal prism.*
- D. *The solid is an octagonal prism.*



Question 49 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

If a prism has a base that is an n -sided polygon, which expression correctly gives the total number of edges of the prism?

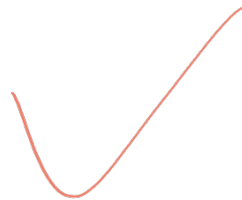
- A. $3n$
- B. $2n$
- C. $n + 2$
- D. $2n + 2$



Question 50 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

How many faces does a pentagonal prism have?

- A. 7
- B. 6
- C. 8
- D. 10



Question 51 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

If the base of a prism is a heptagon, how many faces does the prism have?

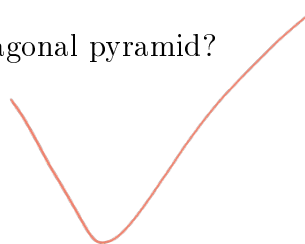
- A. 9
- B. 7
- C. 8
- D. 14



Question 52 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

Which statement is incorrect regarding a pentagonal pyramid?

- A. *It has 10 edges.*
- B. *It has 15 edges.*
- C. *It has 7 faces.*
- D. *It has 5 vertices.*



Question 53 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

What is the number of faces of an octagonal prism?

- A. ☐ 10
- B. ☐ 8
- C. ☐ 9
- D. ☐ 12



Question 54 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 5)

Which statement is true about regular polyhedrons?

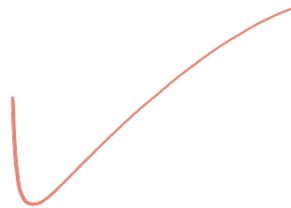
- A. *Only the five Platonic solids are regular polyhedrons.*
- B. *All prisms with regular bases are regular polyhedrons.*
- C. *A pyramid with a regular hexagon base is a regular polyhedron.*
- D. *A regular octahedron can have a square base.*



Question 55 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

How many diagonals does a convex decagon have? Use the formula $d = \frac{n(n-3)}{2}$.

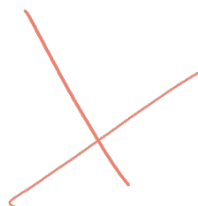
- A. ☐ 35
- B. ☐ 70
- C. ☐ 40
- D. ☐ 50



Question 56 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

If a prism has a base that is a decagon (10-sided polygon), how many edges does it have?

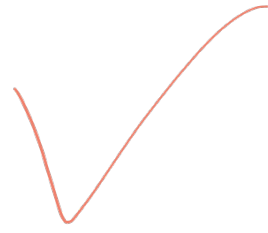
- A. 30
- B. 20
- C. 25
- D. 35



Question 57 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 1)

Which statement is incorrect regarding the number of diagonals in a regular pentagon?

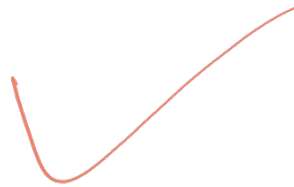
- A. *Theregularpentagonhas5diagonals.*
- B. *Theregularpentagonhas2diagonals.*
- C. *Theregularpentagonhas7diagonals.*
- D. *Theregularpentagonhas10diagonals.*



Question 58 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

Which transformation is equivalent to reflecting a figure over line m and then translating it along line m ?

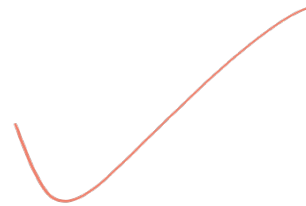
- A. *Aglidereflection*
- B. A rotation about a point on line m
- C. A reflection over line m
- D. A translation along line m



Question 59 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

A regular hexagon allows for rotational symmetries. What are the amplitudes of such rotation symmetries?

- A. 60° , 120° , 180° , 240° , 300° , and 360°
- B. 30° , 60° , 90° , 120° , 150° , and 180°
- C. 60° , 120° , 180° , 240° , and 300°
- D. 60° , 120° , 180° , 240° , 300° , and 420°



Question 60 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

A hexagonal prism has how many faces?

- A. 8
- B. 6
- C. 10
- D. 12



Question 61 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

Consider triangle DEF with angles measuring $(2x + 10)^\circ$, $(3x - 5)^\circ$, and $(x + 25)^\circ$. Determine which angle has the largest amplitude.

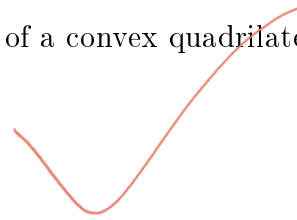
- A. $\angle DEF$
- B. $\angle DFE$
- C. $\angle EDF$
- D. *All angles have the same amplitude.*



Question 62 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

What is the sum of the amplitudes of the exterior angles of a convex quadrilateral?

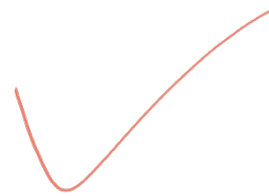
- A. 360°
- B. 180°
- C. 540°
- D. 720°



Question 63 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

In a quadrilateral, the diagonals are equal in length, bisect each other, and are perpendicular. The quadrilateral is a:

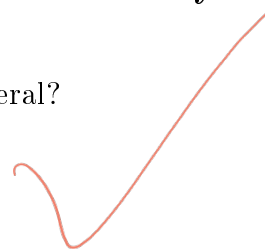
- A. *Square*
- B. *Rectangle*
- C. *Rhombus*
- D. *Isosceles trapezoid*



Question 64 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

What is the sum of the internal angles of a convex quadrilateral?

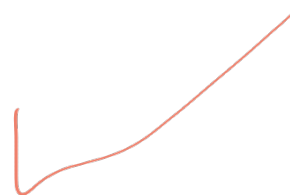
- A. $S_i = (4 - 2) \times 180^\circ = 360^\circ$
- B. $S_i = (4 - 1) \times 180^\circ = 540^\circ$
- C. $S_i = 4 \times 180^\circ = 720^\circ$
- D. $S_i = (4 + 1) \times 180^\circ = 900^\circ$



Question 65 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

Which statement is correct about a prism with a base that is an n -sided polygon?

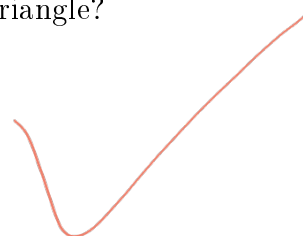
- A. It has $3n$ edges.
- B. It has $2n$ edges.
- C. It has $n + 2$ vertices.
- D. It has $n + 1$ faces.



Question 66 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

What is the sum of the measures of the exterior angles of any triangle?

- A. 360°
- B. 180°
- C. 540°
- D. 720°



Question 67 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 2)

What is the sum of the interior angles of a pentagon?

- A. 540°
- B. 360°
- C. 720°
- D. 450°



Question 68 (Fundamental Mathematics – Elementary Geometry | LLM difficulty: 3)

A hexagonal prism has a base polygon with 6 sides. How many edges does it have?

- A. 18
- B. 12
- C. 14
- D. 16

